EE3001: Solid State Devices
TAs: Alias George (ee24d402@smail.iitm.ac.in) and Devendar K (ee24s015@smail.iitm.ac.in)
Assignment #9

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

If needed, assume the following value: k_BT=0.026 eV (at room temperature)

1. (6 points) A metal-semiconductor Schottky diode (MS1) fabricated on n-GaAs with $N_D = 10^{16}\ cm^{-3}$ has a reverse saturation current density $J_s = 5 \times 10^{-8}$ A/cm^2 at T=300 K. Given $N_c = 4.7 \times 10^{17}\ /cm^3$ and $\epsilon_r = 13.1$ for GaAs.

   (a) Calculate the barrier height (both from metal-to-semiconductor) and semiconductor-to-metal) and the depletion region width at thermal equilibrium assuming the effective Richardson's constant $A^{**} = C_1 N_c/T^2 = 4.4\ A/K^2cm^2$.

   (b) Calculate the minority carrier diffusion current density (assuming this as a p^+n junction) and show that the Schottky diode current density is mostly due to majority carriers (electrons). Assume the hole diffusion coefficient to be 10 cm^2/sec and minority carrier lifetime as 10^{-8} sec.

2. (2 points) We want to design a BJT with β = 150. Consider that for this BJT, base transport factor (α_T) is 0.995. If μ_n/μ_p = 3 and W_B/W_E = 0.5, and all regions are short, what is the required value of N_{DE}/N_{AB}?

3. (2 points) Two transistors T1 and T2 are identical in all respects except in their base widths. The emitter efficiencies are unity in both cases. The base width of T1 is 1 μm and that of T2 is 2 μm. The transistors are biased in the active region of operation. The base current of T1 is 10 μA when the collector current is 1 mA. Determine the base current of T2 when the collector current is 1 mA.

4. (2 points) The β of an n^+pn^+ transistor is 100 when the collector-base junction is reverse biased with $V_{CB} + V_{bi}$ = 1 V. The neutral base width, that is, the undepleted portion of the base for this case is 1.2 μm and the depletion layer width at the CB junction is 0.3 μm. Determine the β of the transistor when $V_{CB} + V_{bi}$ = 9 V considering the following two cases separately: (a) α_T = 1 and $\gamma \neq 1$ (b) $\alpha_T \neq 1$ and γ = 1. Note that V_{CB} is the reverse voltage applied at the CB junction while V_{bi} is the built-in potential of this junction.